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REVIEW ARTICLE



New tools to mitigate drug resistance in *Enterobacteriaceae* – *Escherichia coli* and *Klebsiella pneumoniae*

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ABSTRACT

Treatment to common bacterial infections are becoming ineffective of late, owing to the emergence and dissemination of antibiotic resistance globally. *Escherichia coli* and *Klebsiella pneumoniae* are the most notorious microorganisms and are among the critical priority pathogens listed by WHO in 2017. These pathogens are the predominant cause of sepsis, urinary tract infections (UTIs), pneumonia, meningitis and pyogenic liver abscess. Concern arises due to the resistance of bacteria to most of the beta lactam antibiotics like penicillin, cephalosporin, monobactams and carbapenems, even to the last resort antibiotics like colistin. Preventing influx by modulation of porins, extruding the antibiotics by overexpression of efflux pumps, mutations of drug targets/receptors, biofilm formation, altering the drug molecules and rendering them ineffective are few resistance mechanisms that are adapted by *Enterobacteriaceae* upon exposure to antibiotics. The situation is exacerbated due to the process of horizontal gene transfer (HGT), wherein the genes encoding resistance mechanisms are transferred to the neighbouring bacteria through plasmids/phages/uptake of free DNA. Carbapenemases, other beta lactamases and *mcr* genes coding for colistin resistance are widely disseminated leading to limited/no therapeutic options against those infections. Development of new antibiotics can be viewed as a possible solution but it involves major investment, time and labour despite which, the bacteria can easily adapt to the new antibiotic and evolve resistance in a relatively short time. Targeting the resistance mechanisms can be one feasible alternative to tackle these multidrug resistant (MDR) pathogens. Removal of plasmid (plasmid curing) causing resistance, use of bacteriophages and bacteriotherapy can be other potential approaches to combat infections caused by MDR *E. coli* and *K. pneumoniae*. The present review discusses the efficacies of these therapies in mitigating these infections, which can be potentially used as an adjuvant therapy along with existing antibiotics.

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1. Introduction

Misuse and overuse of antibiotics has taken us from the “golden antibiotic era” to the verge of “post antibiotic era”, where we are left with limited/no therapeutic options available to treat most infectious agents. The emergence and rapid spread of these drug resistant traits in microorganisms has urged WHO to release its first report of List of 12 antimicrobial resistant priority pathogens in 2017, which requires immediate attention from researchers across the globe to address this issue (WHO 2018). *Enterobacteriaceae* are one among the most critical group and WHO has emphasized the research community to develop antibiotics or resistance modulatory approaches to tackle them. The most common and notorious participants of *Enterobacteriaceae* in causing urinary tract infections, blood stream infections

and pneumonia are *Escherichia coli* and *Klebsiella pneumoniae*. Apart from *Acinetobacter baumannii*, these two microbes form the predominant causative agents in neonatal sepsis and owing to their resistance to a broad range of antibiotics, including drugs of last resort, infections caused by these microbes ultimately result in mortality. Nearly, 190,000 children die within four weeks after birth, due to neonatal sepsis in India, out of which, 30% deaths are due to infections by microbes that are multi drug resistant (MDR) (CDDEP 2020). Carbapenems are the antibiotic of choice to treat infections caused by these MDR strains. However, emergence and prevalence of carbapenem resistant *Enterobacteriaceae* (CRE) in the last decade, due to the production of carbapenemases has posed a grave threat to public health. A study performed at Chicago showed a sharp increase in the prevalence of carbapenem intermediate or resistant

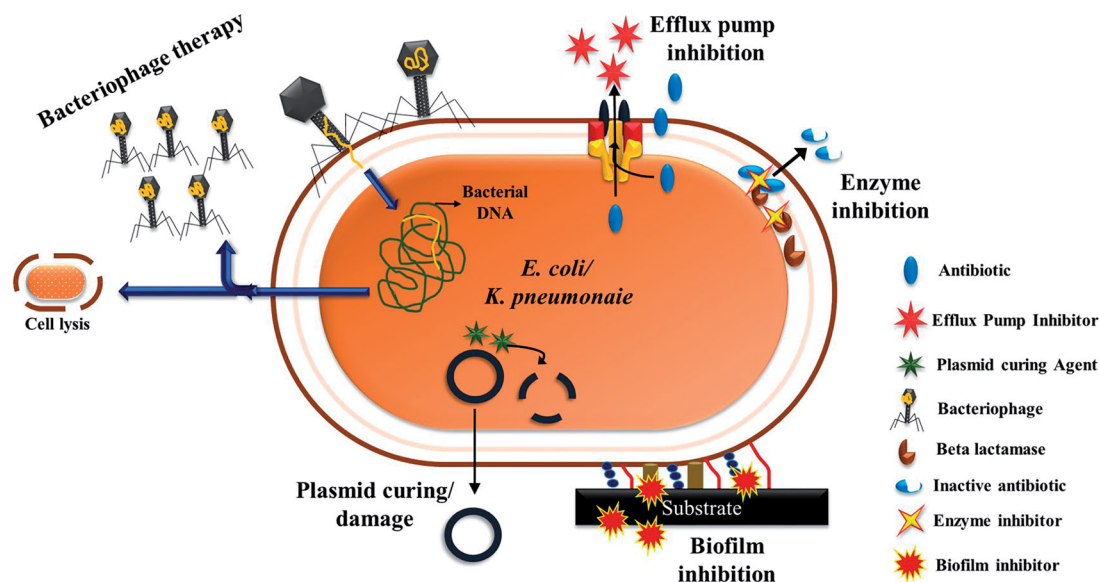


Figure 1. Overview of various strategies that can potentially curtail Multi Drug Resistance in *Enterobacteriaceae*.

Enterobacteriaceae (CIRE/CRE) isolates in patients with the treatment period as 22 CIRE isolates per 1000 patient days (McLaughlin et al. 2013). As per clinical data from 2014, 50% of *K. pneumoniae* from 37% of WHO member states, were found to be carbapenem resistant (WHO 2016). Around 26–44% of deaths were attributed to carbapenem resistance from seven studies as per a meta-analysis, among which, number of deaths among CRE infected patients were twice as that of the infections by carbapenem sensitive *Enterobacteriaceae* (Falagas et al. 2014). Among 120 isolates from stool samples of children less than 5 years at Delhi, India, 47.5% of the isolates showed the presence of *sul1* gene (sulphonamide resistance) and SNP analysis showed mutations in *parC* and *gyrA* gene. These results indicate the development, dissemination and carriage of antibiotic resistant genes in gut microbiota (Singh et al. 2019). Another study showed that commensal *E. coli* was resistant to multiple drugs in all samples from children ($n = 125$) and the isolates from 124 samples produced extended spectrum beta lactamases (Purohit et al. 2019). It is predicted that by 2030, all samples of *E. coli* and *K. pneumoniae* tested in India, will be resistant to last resort drugs (Antibiotic 2020). These studies emphasize the need to control the spread of antibiotic resistance among commensals. Apart from the usual contributors of *Enterobacteriaceae* causing dreadful infections, the SPICE group (*Serratia*, *Providencia*, Indole positive *Proteus*, *Citrobacter* and *Enterobacter* species) of *Enterobacteriaceae* have caused infections with intrinsic resistance to last resort antibiotics (Shalini et al. 2019). High morbidity with mortality, enhanced potential to

cause outbreaks, resistance due to mobile genetic elements, and lack of treatment options are the essential and most compelling rationale that forces us to search for alternate methods to prevent colonization and control infections by these superbugs (World Health Organization 2017). Studies on *E. coli* and *K. pneumoniae* from different regions of the world reveal >60% of these bacteria are resistant to beta lactam antibiotics (Wendt et al. 2010). There are reports where migrants carry and spread resistance to drugs like carbapenems and colistin into low prevalent countries (Soon et al. 2011; Hilty et al. 2012; Von Wintersdorff et al. 2016). Colistin is the drug of last resort for treatment of multi drug resistant *Enterobacteriaceae*. Of late, emergence of both chromosome (alterations in *mgrB* gene, L-Ara-4-N pathway) (Cannatelli et al. 2014; Olaitan et al. 2014) and plasmid mediated resistance (containing genes *mcr-1* to *mcr-9*) (Liu et al. 2016; Yin et al. 2017; Carroll et al. 2019; Vounba et al. 2019; Wang et al. 2019) to colistin has been widely reported. Briefly, the common mechanisms of drug resistance include deficiency or alteration of porins, overexpression of efflux pumps, production of beta lactamases that render the antibiotic ineffective by drug modification, 16s RNA methylases that prevent binding of aminoglycosides, mutation in DNA gyrase and production of pentapeptide Qnr proteins that prevent action of fluoroquinolones (Iredell et al. 2016). This review attempts to cover the possible alternate strategies that has been so far advocated for mitigating multi drug resistant (MDR) *E. coli* and *K. pneumoniae*, which is also graphically represented as Figure 1. A compilation of the most recent reports that target

Table 1. Recent reports targeting different resistance mechanisms against *E. coli* and *K. pneumoniae*.

Mechanism	Targeted bacteria	Compounds	Combination with antibiotic	Reference
Inhibiting Efflux Pumps	MDR <i>E. coli</i> and <i>K. pneumoniae</i>	Ursolic acid	Colistin	(Sundaramoorthy et al. 2019)
	MDR <i>E. coli</i>	Seaweed extracts	Clarithromycin	(Lu et al. 2019)
	Carbapenem and colistin resistant <i>E. coli</i>	Combination of benzyl isothiocyanate, PajN and EDTA and salicylate	Colistin	(Lin et al. 2019)
Plasmid removal	Carbapenem and colistin resistant <i>E. coli</i>	Quercetin		(Sundaramoorthy et al. 2019)
	AcrB overexpressing Carbapenem resistant <i>E. coli</i> and <i>K. pneumoniae</i>	Small molecules	Erythromycin and novobiocin	(Pal and Tripathi 2020)
	Hyperpotentiated <i>E. coli</i> and wild type <i>K. pneumoniae</i>			(Green et al. 2020)
	<i>E. coli</i>	Selenoanhydride and diselenodiester		(Mosolygó et al. 2019)
	<i>E. coli</i>	CRISPR-CAS9 based pFREE		(Lauritsen et al. 2017)
Affecting biofilm formation	<i>E. coli</i>	Ethanol extract of <i>Dioscorea bulbifera</i>		(Temitope Osuntokun et al. 2019)
	<i>E. coli</i>	Interference plasmids		(Kamruzzaman et al. 2017)
	<i>E. coli</i> and <i>K. pneumoniae</i>	Silver nanoparticles	Meropenem and ceftriaxone	(Hashim 2017; Mhawesh et al. 2019)
	MDR <i>E. coli</i> and <i>K. pneumoniae</i>	Pectin capped Platinum nanoparticles		(Bharathan et al. 2019)
	MDR <i>E. coli</i> (Carbapenemase producing)	Abcavir and Azidothymidine		(Buckner et al. 2020)
	<i>E. coli</i> (ESBL producing) and <i>K. pneumoniae</i> (Carbapenemase producing)			
	UPEC	CecropinA (insect antimicrobial peptide)	Nalidixic acid	(Sellera et al. 2019)
	UTI <i>K. pneumoniae</i>	Supernatant of <i>Lactobacillus</i>		(El-Mokhtar et al. 2020; Kheiri et al. 2020)
	Carbapenem resistant <i>K. pneumoniae</i>	Ursolic acid		(Qian et al. 2020)
	MDR <i>K. pneumoniae</i>	Tea polyphenol		(N et al. 2020)
Enzyme Inhibition	MDR <i>K. pneumoniae</i>	Extracts of <i>Allium ursinum</i> and <i>Allium oschaninii</i>		(Galdiero et al. 2020)
	<i>K. pneumoniae</i>	Eugenol		(Qian et al. 2020)
	Carbapenem resistant <i>K. pneumoniae</i>	Low frequency ultrasound	Meropenem, tigecycline, fosfomycin, amikacin and colistin	(Liu et al. 2020)
	<i>K. pneumoniae</i>	Silver nanoparticles		(Foroohmanjili et al. 2020; Siddique et al. 2020)
	<i>K. pneumoniae</i>	Graphene/chitosan nanoparticles		(Muthuchamy et al. 2020)
	UPEC	Chitosan from marine biowaste		(Rubini et al. 2020)
	<i>E. coli</i>	Ceftazidime		(Sun et al. 2020)
	UPEC	Self-assembly by DR15		(Miriyala et al. 2020)
	<i>E. coli</i>	poly [oligo (ethylene glycol) methyl ether methacrylate], poly(MeOEGMA)		(Alves et al. 2020)
	UPEC	Photodynamic therapy in presence of methylene blue		(Gullas et al. 2020)
ROS generation	Carbapenemase producing <i>E. coli</i> and <i>K. pneumoniae</i>	Citric acid, ascorbic acid, malic acid and ciprofloxacin	Imipenem and carbapenem	(Abbas et al. 2019)
	<i>E. coli</i>	CDD-97	Beta lactam antibiotics	(Taylor et al. 2020)
	MDR <i>Enterobacteriaceae</i>	Enmetazobactam		(Tselepis et al. 2020)
	Carbapenemase producing <i>E. coli</i>	Azetidinimide	Imipenem	(Romero et al. 2020)
	<i>E. coli</i>	Ciprofloxacin—pyrrole-2-carboxamide hybrid		(Fois et al. 2020)
	Colistin resistant <i>Enterobacteriaceae</i>	Osthole	Colistin	(Zhou et al. 2019)
	<i>mcr-1</i> positive <i>Enterobacteriaceae</i>	Isoantolactone	Polymyxin B	(Lu et al. 2020)
	<i>mcr-1</i> containing <i>E. coli</i>	1-phenyl-2- (phenylamino) ethanone derivatives	Colistin	(Lan et al. 2019)
	Carbapenemase producing <i>K. pneumoniae</i>	Lavender essential oil	Meropenem	(Yang et al. 2020)
	<i>E. coli</i>	Lactoferrin B like peptide		(Lee et al. 2020)
Altering membrane properties	ESBL producing <i>E. coli</i> and <i>K. pneumoniae</i>	Cinnamaldehyde	Ciprofloxacin and Cefotaxime	(Dhara and Tripathi 2020)
	Carbapenem resistant <i>K. pneumoniae</i>	Hexadecapeptide	Vancomycin and teicoplanin	(Zeng et al. 2020)
	<i>E. coli</i> and <i>K. pneumoniae</i>	Combination of tazobactam and polymyxin B nonapeptide	Piperacillin	(Ferrer-Espada et al. 2020)
	MDR <i>Enterobacteriaceae</i>	Dilipid ultrashort tetrabasic peptidomimetic	Novobiocin and Rifampicin	(Ramirez et al. 2020)
	MDR <i>K. pneumoniae</i>	Molecular nanomachine	Meropenem	(Galbadage et al. 2019)

different mechanisms leading to drug resistance in *Enterobacteriaceae* (Table 1) has also been presented.

2. Drug resistant Bacteria- to kill or to disarm?

Most of the antibiotics like cephalosporins, penicillin, quinolones and macrolides were discovered and developed in the 20th century, with 1950–1960 being the golden era of antibiotics (Fair and Tor 2014). However, off late, there are only a minimal number of new antibiotics being developed, despite the urgent necessity. The pharmaceutical companies have minimized their investment in discovery of new antibiotics. The major reason for this is due to the rapid evolution of bacterial resistance to antibiotics. Overuse and misuse of antibiotics in agriculture, aquaculture, poultry, and indiscriminate sale of over the counter antibiotics are few common human mediated factors that has contributed to the increased prevalence of antibiotic resistant bacteria. Presence of resistance traits in the bacterial genome were observed even from permafrost samples that were 30,000 year old, underscoring inherent existence of antibiotic resistant genes in evolution (as a part of niche competition), long before the discovery of man-made/synthetic antibiotics (D'Costa et al. 2011). Antibiotics usually cause bacteriostatic or bactericidal effect, which exerts selective pressure on bacteria to survive, thus resulting in evolution of drug resistant bacteria through adaptive mutations/by acquisition of resistant traits through horizontal gene transfer (Totsika 2017). In addition, most antibiotics exert broad spectrum effect and eliminate commensal bacteria leading to dysbiosis, which in itself is a cause for dreadful infections like systemic candidiasis/pseudomembrane colitis (Mason et al. 2012; Farooq et al. 2015). Thus efforts to develop novel antibiotics that kill bacteria will end up in vain and shall not provide a tangible solution to tackle these MDR pathogens, instead it will give rise to an altered gut microbiome and a new pile of pathogens with resistance to the novel antibiotics. Hence, in order to mitigate them effectively without further burdening the globe with antibiotic resistant pathogens, alternate approaches that would disarm or diffuse the resistance mechanisms of bacteria can become a promising solution. Unlike antibiotics, the resistance modulatory approach of targeting the resistant factors viz., efflux pump inhibitors, plasmid curing agents, quorum quenchers etc., can abrogate the evolution of resistant bacteria. One other advantage of the alternate (resistance modulatory) approach is that we can have multiple targets to inhibit. They can be customized in most of the cases such that they target only the pathogenic

strains and not the commensals. Few of the resistance modulatory approach like inhibiting efflux pumps, plasmid curing, enzyme inhibition have been broadly discussed in the current review. Other alternate approaches like use of bacteriophages, bacteriotherapy has also been discussed.

3. Targeting resistance-causing mechanisms

3.1. Efflux pump inhibition

Overexpression of efflux pumps like AcrAB-TolC, MdfA, MacA, MacB, KpnEF, KdeA, EmrKY-TolC and EmrAB-TolC leads to decreased susceptibility to diverse antibiotics by reduction in intracellular antibiotic concentration (Nikaido and Pagès 2012; Lamut et al. 2019). In addition, expression of efflux pump genes reduces the susceptibility of *K. pneumoniae* to biocides like chlorhexidine, benzalkonium chloride and Triclosan (Abuzaid et al. 2012). AcrAB-TolC system is a predominant RND (Resistance nodulation division) class of efflux pump and it is the major contributor of multidrug resistance with aminoglycosides, tetracyclines and carbapenems as their efflux substrates (Chetri et al. 2019). 3-amino-6-carboxyl-indole and 3-nitro-6-amino indole exhibited synergy with chloramphenicol, tetracycline, erythromycin and ciprofloxacin against *E. coli* reducing their minimum inhibitory concentration (MIC) by 2–64 fold (Zeng et al. 2010). In addition, the indole derivatives were able to inhibit TolC component of efflux pump, thus resulting in enhanced intracellular concentration of the antibiotics. Pyranopyridine-based derivatives were found to inhibit the soluble periplasmic part of AcrAB-TolC pump (AcrB) as evidenced by molecular studies (Sjuts et al. 2016). *In silico* virtual screening suggested 2 best inhibitors among the phytochemical compounds—lanatoside C and diadzein, which caused enhanced accumulation of ethidium bromide (EtBr) *in vitro* in *E. coli* (Aparna et al. 2014). Another study showed that derivatives of ursolic acid – 3-O-acetyl-urs-12-en-28-isopropyl ester (UA-4) and 3-O-acetyl-urs-12-en-28-n-butyl ester (UA-5) reversed the MIC of nalidixic acid by 8 fold in nalidixic acid resistant *E. coli* and exhibited synergy with tetracycline against MDR clinical isolate of *E. coli* by inhibiting efflux pump (Dwivedi et al. 2015). Our group has recently showed that ursolic acid was able to inhibit efflux pump and resensitized colistin resistant MDR clinical isolates of *E. coli* and *K. pneumoniae* by synergistic interaction with colistin (Sundaramoorthy et al. 2019). Ursolic acid was shown to enhance the bactericidal activity of colistin by 4–4.5 log fold *in vitro* time kill assay and reduced the bioburden by 1–1.58 fold *in vivo* zebra fish infection

(Sundaramoorthy et al. 2019). Quinazoline derivatives increased the activity of chloramphenicol and nalidixic acid, by enhancing the intracellular drug concentration in AcrAB-TolC pump overexpressing *K. pneumoniae* strains (Chevalier et al. 2010). 2,8-dimethyl-4-(2'-pyrrolidinoethyl)-oxyquinoline, an alkoxyquinoline derivative, restored drug susceptibility by inhibiting chloramphenicol efflux in MDR *K. pneumoniae* isolates (Chevalier et al. 2004). A recent study showed that extracts of red and brown seaweeds containing terpenes, phenolic compounds, alkaloids and halogenated compounds, enhanced accumulation of EtBr and was synergistic with clarithromycin against drug resistant *E. coli* (Lu et al. 2019). Combination of benzyl isothiocyanate (BITC) with EPIs (CCCP—carbonyl cyanide *m*-chlorophenyl hydrazine and PaβN—phenylalanine-arginine β-naphthylamide) and metal chelator (ethylene diamine tetra acetic acid—EDTA) was evaluated against MDR *E. coli* (Lin et al. 2019). Double and triple combinations of BITC, EDTA and PaβN showed synergistic effect, which was reflected in the time kill assay, whereas combination with CCCP showed additive/indifference effect (Lin et al. 2019). We have shown that a combination of MarR inhibitor—salicylate, in combination with an efflux pump inhibitor benzochromene derivative BC1 significantly reduced the MIC of colistin against colistin resistant clinical isolate of uropathogenic *E. coli* (Sundaramoorthy et al. 2019). The triad combination of Colistin + BC1 + Salicylate caused a 6 log and 4 log reduction in CFU/mL *in vitro* time kill study and *in vivo* infection model respectively (Sundaramoorthy et al. 2019). A follow up study with the synthetic EPI compound (benzochromenes) replaced with a plant nutraceutical curcumin, also inhibited efflux of colistin and potentiated its activity against colistin resistant clinical isolates of *E. coli* and *K. pneumoniae* (Sundaramoorthy et al. 2020). 4-hydroxy- α -tetralone and its derivative reversed MIC of tetracycline against MDR *E. coli* (Dwivedi et al. 2014). Mechanistic studies revealed that the compounds inhibited ATP dependent efflux and downregulated *yojI* efflux pump gene (Dwivedi et al. 2014). Another study showed the biphenylmethylene and fluorenylmethylene derivatives of piperazine arylideno-imidazolones inhibited efflux in AcrAB-TolC overexpressing strain of *E. coli* and reduced the MICs of oxacillin, linezolid, levofloxacin, rifampin and clarithromycin by 2–32 fold. In addition to their EPI activity, they were found to permeabilize outer membrane as evidenced by the nitrocefin hydrolysis assay (Bohnert et al. 2016). A pyranopyridine compound, MBX2319, reduced the MIC of ciprofloxacin,

levofloxacin and piperacillin by 2–8 fold and enhanced accumulation of Hoechst 33342 in *E. coli*, but not in AcrAB-TolC knocked out strain, thus inhibiting efflux of AcrAB-TolC component (Opperman et al. 2014). A recent study showed that quercetin inhibited efflux in *acrB* overexpressing carbapenem resistant *E. coli* and *K. pneumoniae*, owing to the adequate release of Gibbs' free energy and involvement of identical set of amino acids for interaction with the pump as observed with PAβN (Pal and Tripathi, 2020). Six small molecules with a shared chemical scaffold and substituted with 4(3-aminocyclobutyl)pyrimidin-2-amine were shortlisted from docking based on physicochemical properties. They were found to inhibit AcrAB-TolC of *E. coli* and enhanced the activity of erythromycin and novobiocin against hyperporinated *E. coli* and wild type strains of *K. pneumoniae* (Green et al. 2020). Selenocompounds—selenoanhydride and diselenodiester were found to actively inhibit AcrAB-TolC pump of *E. coli* (Mosolygó et al. 2019).

3.2. Plasmid curing

The resistance contributing factors can be either chromosomally mediated or plasmid mediated. Plasmids in the form of accessory genome (60–200 kb) comprise 10% of total genome of *E. coli* and *K. pneumoniae* and contribute to dissemination of antibiotic resistance (Sherley et al. 2003). Involvement of conjugative plasmids have worsened the resistance condition as they can convert antibiotic susceptible strains to multi drug resistant virulent strains after one-two doses of antibiotics (van Hal et al. 2009). Few detergents like sodium dodecyl sulphate (SDS), bile and intercalating agents like acridine orange and ethidium bromide are well known for their plasmid curing ability in *E. coli* and *K. pneumoniae* (Zaman et al. 1970; Salisbury et al. 1972; Mansi et al. 2000). Nevertheless, their adverse effects and requirement of higher concentrations hinder their use in practice (Michelle et al. 2018). Antagonism of the B subunit of DNA gyrase by semi-permissive concentrations of coumermycin A1, eliminated plasmids from *E. coli* (Wolfson et al. 1982). An universal plasmid curing system (pFREE) was constructed based on CRISPR—CAS9 targeting, which achieved curing efficiencies between 40 and 100% in *E. coli* (Lauritsen et al. 2017). Another CRISPR-Cas9 mediated plasmid curing system—pCasCure was developed, which cured *bla*_{KPC} harbouring IncFIIK-pKpQIL and IncN pKp58_N, *bla*_{NDM} harbouring IncX3 plasmid and *bla*_{OXA-48} harbouring pOXA-48-like plasmids from *Enterobacteriaceae* (Hao et al. 2020). Continuous exposure to irgasan (an

antimicrobial compound) impregnated interpenetrating polymer network (IPN) silicone hydrogel, caused 83% plasmid loss in *E. coli* in a liquid suspension (Riber et al. 2016). 8-epidiosbulbin E acetate (norditerpene) extracted from bulbs of *Dioscorea bulbifera* L., cured plasmids from clinical isolates of *E. coli* with 12–48% curing efficiency and did not pose any toxicity as evidenced by the cell line studies (Shriram et al. 2008). Another recent study showed that ethanolic extract from tubers of *Dioscorea bulbifera* contained bioactive compounds saponin, tannin, flavonoids, terpenoid, phlobatannin and cardiac glycosides and eliminated plasmid, caused potassium ion leakage in *E. coli* (Temitope Osuntokun et al. 2019). The antibiotic, rifampicin, cured F^+ cells from *E. coli* episome at low dose (Bazzicalupo and Tocchini-Valentini 1972). Small molecules were developed based on fragment-based screening, which can inhibit dimerization of TraE (conjugal transfer protein), thus inhibiting transfer of plasmid by conjugation (Casu et al. 2017). Psychotropic drugs, amitriptyline and 7,8 dioxochlorpromazine inhibited conjugal transfer of resistant plasmid in *E. coli* (Molnár et al. 1980). Promethazine, 9-aminoacridine and trifluoperazine exhibited low plasmid curing effect, despite forming complex with plasmid DNA of *E. coli* (Spengler et al. 2003). The efficiency was enhanced when combined with calmodulin and proton pump inhibitors. Combination of trifluoromethyl-ketone with calmodulin antagonist promethazine or 9-amino acridine promoted the plasmid eliminating capacity in *E. coli* (Wolfart et al. 2006). Plasmid (Pili encoded by F factor) dependent phage PRD1, reduced the frequency of antibiotic resistant strains of *E. coli* (Jalasvuori et al. 2011). This occurs due to the evolution of phage resistant but antibiotic susceptible strains, which were not able to conjugate (Jalasvuori et al. 2011). Interference plasmids were constructed by deletion of toxins and antibiotic resistance genes from the target plasmid, which cured the respective target plasmid from *E. coli* and *K. pneumoniae* *in vitro*, restoring antibiotic susceptibility *in vivo* (Kamruzzaman et al. 2017). SILA compound – 1,3-dimethyl-1,3-bis(4-fluorophenyl)-1,3-bis{3-[1(4-buthyl-piperazinyl)]-propyl}-disiloxantetrahydrochloride (SILA 421) reduced the load of plasmid by 5 log fold in efflux pump overexpressing *E. coli* and reversed the resistance of the strains to tetracycline (SCHELZ et al. 2007). Plasmid cured variants of *E. coli* strains that carried ESBL plasmids with toxin-antitoxin systems were constructed by using relatively high temperature (45 °C) in brain heart infusion broth (Schauffler et al. 2013). Few studies have evaluated the plasmid curing ability of nanoparticles. Silver nanoparticles cured plasmid from MDR *E. coli* strains in a dose

dependent manner (0.75 ppm–3 ppm) resulting in a parallel increase in curing efficiency (50–100%) (Mhawesh et al. 2019). Another study showed the ability of silver nanoparticles, synthesized from aqueous extract of *Zingiber officinale*, to eliminate plasmids from *E. coli* and *K. pneumoniae* (Hashim 2017). We observed that Pectin capped platinum nanoparticles could cure IncF type plasmid bearing carbapenamses from MDR *E. coli* and reduced the MIC of meropenem and ceftriaxone by 16–64 fold (Bharathan et al. 2019). The plasmid cured strain possessed a small colony phenotype and displayed 50% reduction in biofilm formation relative to its wild type. A curing efficiency of 17% was achieved *in vivo* in a zebra fish infection model and the biogenic platinum nanoparticle reduced the bioburden by 2.4 log, when used as an adjuvant along with meropenem (Bharathan et al. 2019). Mechanistic studies showed that the nanoparticles disrupted the membrane integrity and caused DNA cleavage. A recent study screened Prestwick library of compounds and identified that HIV drugs—abacavir and azidothymidine reduced transfer of plasmids in *E. coli* (pCT—ESBL producing plasmid) by 83.3% and in *K. pneumoniae* (pKpQIL—carbapenemase producing plasmid) by 80.8% at reduced concentrations that can be achieved in the gastrointestinal tract (Buckner et al. 2020).

3.3. Enzyme inhibition

Enzymes altering the drugs include beta lactamases, broad-spectrum beta lactamases, extended spectrum beta lactamases (ESBLs) and carbapenemases. These enzymes hydrolyse the compounds containing beta lactam ring (penicillin, cephalosporins and carbapenems) either by formation of acyl enzyme by active serine or by one/two essential zinc ions in the active sites (Bush 2018). Quercetin exhibited synergy with meropenem and reduced its MIC upto 4 fold against CRE (Pal and Tripathi, 2020). In addition, quercetin altered the expression of *bla_{VIM}* and *ompC* expression. Pterostilbene exhibited synergy with polymyxin B and the combination eradicated *mcr-1* and *NDM-1* positive *E. coli* strain *in vitro* 1 h post inoculation (Zhou et al. 2018). Citric acid, ascorbic acid, malic acid and ciprofloxacin exhibited synergy with imipenem/meropenem at their sub-inhibitory concentrations against carbapenemase producing *E. coli* and *K. pneumoniae* isolates. The activity can be due to chelation of zinc ions in the active site of carbapenemase (Abbas et al. 2019). NXL104 (Avibactam sodium) inhibited KPC-2 (IC₅₀), thus sensitizing drug resistant KPC-2/KPC-3 producing *Enterobacteriaceae* strains to ceftriaxone, imipenem,

piperacillin and reduced the MIC of ceftazidime by 1000 fold at 4 mg/L (Se Stachyra et al. 2009). A serine beta lactamase inhibitor, vaborbactam, restored the sensitivity of meropenem in 72.7–98.1% out of 315 serine carbapenemase producing *Enterobacteriaceae* isolates at 8 mg/L (Castanheira et al. 2016). Another study discovered a serine β lactamase inhibitor, RPX7009 with a boronic acid pharmacophore (Hecker et al. 2015), which reduced the MIC of cefepime, biapenem, meropenem, ertapenem and imipenem by 4–130 fold against different clinical strains of serine carbapenemases producing *Enterobacteriaceae*. RPX7009 in combination with meropenem decreased 2 log CFU/mL of KPC producing *K. pneumoniae* infection in a neutropenic mouse lung infection model (Hecker et al. 2015). Bisthiazolidine (BTZ) with features of β lactams that can be altered with metal binding groups was designed to target the active site of metal β lactamases (MBLs) like NDM-1 [81]. Hydrolysis of imipenem was prevented by BTZs in NDM-1- producing *E. coli*, as evidenced by NMR spectroscopy (González et al. 2015). DNA encoded chemical library approach shortlisted a compound, CDD-97, as an inhibitor of oxacillinase-48 (OXA-48), binding non-covalently to active site of the enzyme and structure activity relationships showed a two-fold increase in potency (Taylor et al. 2020). Unexpectedly, the compound exhibited poor synergy with β -lactam antibiotics against *E. coli*, owing to its poor accumulation (Taylor et al. 2020). Relebactam, a compound with diazabicyclooctane core with piperidine side chain, retained the susceptibility of imipenem against different clinical isolates of KPC producing *Enterobacteriaceae*, by inhibiting KPC-2 lactamase (Papp-Wallace et al. 2018). The KPC-2-relebactam complex was stable for 24 h as revealed by the time—based mass spectrometry (Papp-Wallace et al. 2018). A recent study showed that, AAI-101 (enmetazobactam) effectively inhibited class A β -lactamases in MDR *Enterobacteriaceae* but could not inhibit KPC and OXA-48 enzymes (Tselepis et al. 2020). An azetidinimine compound, 7dfm, inhibited KPC-2, OXA-48 and NDM-1, despite the structural differences among the enzymes, at submicromolar level and sensitized NDM-1 expressing *E. coli* to imipenem (Romero et al. 2020). Complexation of zinc ions in NDM-1 or hydrogen bonds in KPC-2 and OXA-48 increases the affinity of compound by allocating the aryl groups in the enzyme pocket (Romero et al. 2020). Hybrids of GyrA (ciprofloxacin) and GyrB inhibitors (pyrrole-2-carboxamide) were synthesized with an intention to inhibit gyrase in *E. coli*, along with delaying the onset of resistance in bacteria (Fois et al. 2020). Entry of the hybrid into the cell shall be taken care by ciprofloxacin component, a

highly permeable GyrA inhibitor. Despite the novelty, the hybrids did not potentiate the antibacterial activity and they lost their antibacterial activity in a GyrA mutant, indicating the interaction of the hybrid occurs with the GyrA subunit and not with GyrB (Fois et al. 2020). This could be overcome by forming a hybrid with benzothiazole GyrB inhibitor with ciprofloxacin connected with a cleavable linker could result in strong inhibition of both sub units. A study showed that peptide conjugated phosphorodiamidate morpholino oligomers (PPMOs) targeting *mcr-1* mRNA were able to restore polymyxin sensitivity in *mcr-1* positive *E. coli* strains and reduced the MIC by 2 to 16 fold (Daly et al. 2017). This antisense approach in combination with colistin, reduced morbidity and bacterial burden in spleen at 24 h in a sepsis BALB/c mice model (Daly et al. 2017). Osthole (OST), by its synergistic interaction with colistin, eliminated *mcr-1* positive *Enterobacteriaceae* in 3–7 h and reduced *in vivo* bacterial load in a mouse infection model (Zhou et al. 2019). Isoalantolactone sensitized *mcr-1* positive *Enterobacteriaceae* to polymyxin B, causing a significant decline in bacterial growth *in vitro* and mice infection model (Lu et al. 2020). Derivatives of 1-phenyl-2- (phenylamino) ethanone (6p and 6q) at 25 μ M completely inhibited the growth of *mcr-1* containing *E. coli* strain in combination with colistin (2 μ g/mL) (Lan et al. 2019). Enzymatic study revealed that the compounds inhibited the activity of MCR-1 protein by interacting with Glu246 and Thr285 residues of the protein as evidenced by the docking study.

3.4. Inhibiting the protein secretion systems

Secretion of proteins by bacteria play a predominant role in pathogenic connotation with the host organisms. Gram-negative bacteria possess single step protein secretions systems that transport substrates from bacterial cytoplasm to target cell or to extracellular space, namely, Type I (T1SS), Type III (T3SS), Type IV (T4SS) and Type VI (T6SS) that span both inner (IM) and outer membrane (OM) (Tseng et al. 2009; Costa et al. 2015). The secretion systems aid in transport of hydrolysing enzymes, toxins or effector proteins, which promotes bacterial invasion and colonization and they also act as mediators for conjugation of plasmid DNA that in turn results in spread of antimicrobial resistance genes (Costa et al. 2015). The secreted proteins are known to play a major role in enhancing the virulence of bacteria by facilitating attachment to eukaryotic cells, scavenging energy sources and intoxicating the target cells. Hence, inhibiting the protein secretion systems can be

one of the potential targets to tackle multidrug resistant pathogenic bacteria. Among the different secretion systems, T3SS and T6SS has been reported widely in contributing to virulence in *E. coli* and *K. pneumoniae* by secretion of effector proteins and T4SS by playing a role in bacterial conjugation. In *E. coli*, compounds like rottlerin (plant derived), heterocyclic 2- pyridines, thia-diazolidine- 3,5—diones, derivatives of 8—aminoimidazo[1,2-a]pyrazine have been reported to inhibit T4SS mediated conjugal transfer either by disrupting T4SS apparatus biogenesis or by inhibiting VirB11/TrwD ATPase activity, which are essential to energize the bacteria throughout the process of conjugation (Boudaher and Shaffer 2019). A natural compound from *Streptomyces goldiniensis*, Aurodox, was able to inhibit T3SS by downregulating 25 genes related to T3SS in enteropathogenic and enterohemorrhagic *E. coli* and hence reduced its infecting ability in epithelial cells (McHugh et al. 2019). Guadinomines A and B, isolated from *Streptomyces* sp., showed inhibition of T3SS mediated haemolysis of sheep blood cells by enteropathogenic *E. coli* with IC₅₀ of 0.02 and 0.007 µg/mL, respectively (Iwatsuki et al. 2008). A study showed that the anti-nanobody to Tir (translocated intimin receptor—first effector protein translocated by T3SS), TD4, inhibited the attachment of EHEC to HeLa cells and decreased the adherence to colonic mucosa (Hotinger et al. 2021). A derivative of halogenated salicylaldehyde condensed with 3—aminoacetophenone derivative, Compound 1, decreased the level of type III secreted proteins—EspB and Tir inside EPEC, with IC₅₀ of 15 µM (Gauthier et al. 2005). Apart from T3SS inhibitors for *E. coli*, recent studies have shown that compounds like myricetin, sanguinarine chloride, cytosporone B, 4EpDN (a member of cyclic peptomer) efficiently inhibited T3SS of other members of *Enterobacteriaceae* like *Salmonella* sp, *Yersinia* sp (Hussain et al. 2021; Lam et al. 2021; Lv et al. 2021). A recent study designed a biomimetic cyclic peptide (BCP) specific to a protein-protein interface, TssK-TssG, which is one of the key factors in assembly of T6SS baseplate (Cherrak et al. 2021). BCP was able to prevent the assembly of EAEC—T6SS base plate complex and thus reducing the virulence of the pathogenic bacteria during host invasion. Another study developed two nanobodies—nb02 and nb25, specific to TssM of EAEC by immunization of Llamas, out of which nb25 disrupted the interaction of TssM-TssJ, thus interfering in proper function of T6SS apparatus (Nguyen et al. 2015). Most of the secretion system inhibitors reported have shown to affect the virulence and its associated genes of the bacteria but not the essential genes and does not predominantly interfere

with the bacterial viability, thus avoiding the selective pressure to develop resistance against these inhibitors.

4. Biofilm inhibition

The common virulence factors of *Enterobacteriaceae* include adhesions, haemolysin production and biofilm formation (Lüthje and Brauner 2014). Among these, biofilm-forming property is one of the predominant reason leading to recurrent urinary tract infections (UTI) by *Enterobacteriaceae*. In addition, bacterial biofilms in hospital settings enhances the probability of nosocomial infection and can increase the frequency of horizontal gene transfer (HGT), thus resulting in acquisition of antibiotic resistant genes (Vuotto et al. 2017). Studies on clinical samples have shown that most of the uropathogens are strong biofilm formers and it is essential to consider this property while treating UTI (Pradhan et al. 2019; Whelan et al. 2020). A recent study showed that biofilm formation is essential for MDR *K. pneumoniae* to adhere to the intestine and colonies in solid organ transplant patients (Ramos-Vivas et al. 2019). Another study showed the destruction of uropathogenic *E. coli* (UPEC) biofilms by an insect antimicrobial peptide, cecropinA (CecA) (Sellera et al. 2019), which exhibited synergism with nalidixic acid and permeabilized the outer membrane of UPEC, ultimately leading to cell death. Addition of 2 mM ethylenediamine tetraacetic acid (EDTA) reduced the MIC of meropenem and nitrofurantoin by 81% and 61%, respectively, in uropathogenic *E. coli* (Wesam et al. 2017). Gelatin-EDTA coating inhibited biofilm formation by 45–81% at 10 – 20 mM (Wesam et al. 2017). Supernatants of *Lactobacillus* possessed 95% biofilm inhibition property on strong biofilm forming UTI isolate of *K. pneumoniae* at 1:1 to 1:16 dilutions (Kheiri et al. 2020). Another similar study showed that cell free supernatants of *Lactobacilli* inhibited young biofilms and disrupted preformed biofilms of ESBL producing *K. pneumoniae* (El-Mokhtar et al. 2020). Metabolites or extracts from plants have been reported widely for their efficient anti-biofilm activity. Ursolic acid was found to inhibit biofilm formed by carbapenem resistant *K. pneumoniae* and inactivated the cells encased in biofilm (Qian et al. 2020). Sub inhibitory concentration of tea polyphenols inhibited MDR *K. pneumoniae* biofilm and slime formation (N et al. 2020). Clove oil decreased the extracellular protein and polysaccharide content by 62% and 47%, respectively and when formed as solid liposomes, it inhibited biofilms of *E. coli* O157:H7 at both 4 °C and 25 °C (Cui et al. 2020). Extracts of *Allium ursinum* and *Allium oschaninii* inhibited biofilms (static and dynamic model) of *K. pneumoniae* *in vitro* (Galdiero et al. 2020). Interestingly,

biofilms formed on Teflon were more susceptible to the extracts than biofilms formed on polycarbonate polypropylene (Galdiero et al. 2020). Herbal *Berberis aristata*, exhibited 51% biofilm inhibitory action and 91% anti-adhesion property against carbapenem resistant *E. coli* (Thakur et al. 2016). Grape seed extract (GSE) inhibited biofilm produced by CRE at 2000, 4000 and 8000 µg/mL after 72 h (Farshkhahi et al. 2018). Eugenol inhibited biofilm formation and inactivated cells in biofilm of carbapenem resistant *K. pneumoniae* at 2X MIC, moreover, it downregulated four biofilm related genes—*luxS*, *wbbM*, *wzm* and *pgaA* (Qian et al. 2020). Coating urinary catheter with a self-assembled layer formed by a derivative of DR15 with linseed oil, restricted biofilm formation when incubated with artificial urine (Miryala et al. 2020). Chloroform extracts of ginger (*Zingiber officinale*) and nutmeg (*Myristica fragrans*) exhibited biofilm inhibition activity on *E. coli* at different concentrations with maximum efficiency (Challaraj Emmanuel et al. 2020). Combination of catechin, protocatechuic and vanillic acid reduced 26–51% of motility and 20–40% of carbohydrate and protein content in biofilm matrix of uropathogenic *E. coli* on silicone catheters (Bernal-Mercado et al. 2020). In addition, surface charge, curli fimbriae and cell hydrophobicity was much altered by the combination. Vanillic acid contributed to reduced cell adhesion, reduced EPS production and motility, whereas protocatechuic acid and catechin altered cell surface property and membrane functionality, respectively (Bernal-Mercado et al. 2020). Another study showed that thymol inhibited biofilm formed by ESBL producing *E. coli* at its sub MIC (50–200 µg/mL) (Cusimano et al. 2020). A polymer brush—poly [oligo (ethylene glycol) methyl ether methacrylate], poly(MeOEGMA) reduced 57% surface area covered by biofilm formed by *E. coli* in a parallel plate flow chamber (PPFC) under hydrodynamic conditions mimicking the urinary catheters and stents (Alves et al. 2020). Further, antibiotic treatment potentiated cell death by reducing the total cell number by 88% (Alves et al. 2020). Nanoparticles have also been reported for their potential biofilm inhibitory activity. Silver nanoparticles synthesized from extract of *Mespilus germanica*, disrupted preformed biofilm and inhibited biofilm formation by *K. pneumoniae*, in addition, the expression of *mrkA* gene (type 3 fimbriae) and *luxS* gene was also downregulated (Foroohimanjili et al. 2020). Silver nanoparticles synthesized from extract of *Nitzschia palea*, reduced curli and inhibited biofilm formed by ESBL producing *E. coli* at 300 ng/mL (Shafreen et al. 2017). Another study showed that silver nanoparticles decreased the production of extracellular polymeric substances and disrupted cell membrane. AgNPs inhibited

64% and 86% of biofilms formed by two different MDR strains of *K. pneumoniae* (Siddique et al. 2020). Graphene/chitosan nanoparticles inhibited biofilm formation of *K. pneumoniae* by 92% at 40 µg/mL and caused a damage to cell surface as evidenced by scanning electron microscopy (SEM) (Muthuchamy et al. 2020). Chitosan extracted from marine biowaste was found to curb 80–85% of quorum sensing mediated biofilm formation in uropathogenic *E. coli*. It also suppressed swarming motility, reduced cellulose production and downregulated invasion related virulent genes (Rubini et al. 2020). Paeoniflorin exhibited antibiofilm activity and inactivated cells within biofilms of carbapenem resistant *K. pneumoniae* (Qian et al. 2020). Initial addition of 4% trisodium citrate (TSC) to *K. pneumoniae* strains resulted in significant reduction of biofilm on day 4, relative to addition of TSC post biofilm formation on day 3 (Chu et al. 2020). A multi drug and carbapenem resistant *Enterobacteriaceae* strain producing beta lactamase, was found to be susceptible to eravacycline (Zhang et al. 2016). It was observed that eravacycline could inhibit biofilm formed by uropathogenic *E. coli* at 0.5 µg/mL, whereas other antibiotics like colistin and meropenem could eradicate only at 32 µg/mL and 2 µg/mL, respectively (Grossman et al. 2015). Self-assembled monolayers (SAM) of alkanethiols functionalized with tri(ethylene glycol) on gold films resisted *E. coli* biofilm upto 99.5% for 1 day (Hou et al. 2007). Ceftazidime inhibited biofilm formation, swimming motility of *E. coli* at sub-MIC by downregulating the *ibpA* gene and increased the extracellular indole concentration by upregulating *tnaA* (tryptophanase) gene (Sun et al. 2020). The compounds, 1-Amino-4-hydroxyanthraquinone (Disperse Red 15 or DR15) and 4-(4'-chloro-4-biphenyl)sulfonylamino benzoic acid (CB1), inhibited biofilm formation, auto aggregation and swarming motilities of uropathogenic *E. coli* when subjected to photodynamic therapy in presence of photosensitizer—methylene blue (2.5X), a 3 log reduction in biofilm colony counts and enhanced the susceptibility of the strains to antibiotics were observed (Gulías et al. 2020). Low frequency ultrasound (92.36mW/cm²) enhanced the bactericidal activity of meropenem, tigecycline, fosfomycin, amikacin and colistin against biofilm infection caused by *K. pneumoniae* and the ultrasound treatment exhibited synergy with the antibiotics upon either single (5 min) or multiple exposure (5 min to 8 h) (Liu et al. 2020).

5. Use of bacteriophages

Bacteriophages (bacterium eater), coined by a Franco-Canadian microbiologist—Felix d'Herelle, represent the

viruses that infect bacterium and follow either a lytic or a lysogenic phase (Fruciano et al. 2007). Phage therapy, initiated in 1900s, with first human therapy in 1920s, was effective against different pathogens, which faced a downfall after the emergence of antibiotics (Gordillo Altamirano and Barr 2019). Owing to emergence of multidrug resistant traits and lack of treatment options available, phage therapy resurged again. Pyophage (PYO)—targeting *S. aureus*, *E. coli*, *P. aeruginosa* and Inestiphage (INTESTI) – 23 different enteric bacteria are products of The Eliava institute of Bacteriophage, Microbiology and Virology, Georgia that are marketed for pyoinflammatory and enteric diseases (Eliava 2008). A study showed that lytic bacteriophage belonging to family *Podoviridae*, ϕ BO1E, targeting KPC producing *K. pneumoniae* strains of clade II lineage, was isolated, which protected larvae of *Galleria monella* infected with KPC producing and colistin resistant *K. pneumoniae* (D'Andrea et al. 2017). Commercial phage cocktails—PYO, INTESTI and Septaphage made *E. coli* susceptible with efficiency 61%, 67% and 9%, respectively (Bernasconi et al. 2017). In addition, five strains of carbapenemase producing *E. coli* and three colistin resistant *E. coli* were found to be susceptible to PYO and INTESTI (Bernasconi et al. 2017). Bacteriophages against *E. coli*, *Enterobacter* species and *K. pneumoniae* were isolated from sewage samples belonging to *Phieco32virus*, *Myoviridae* and *Podoviridae*, respectively (Manohar et al. 2019). Cocktail of the phages reduced the bacterial load by 3 log in 2 h and were also found to effective against meropenem and colistin resistant strains (Manohar et al. 2019). Bacteriophage, vB_KleS_HSE3, belonging to a novel phage lineage of *Siphoviridae*, was isolated from hospital sewage that infected four multi drug resistant *K. pneumoniae* strains with a large burst size of 277 PFU/bacterial cell (Peng et al. 2020). A recent study showed that a custom-made bacteriophage preparation was administered as oral and intra-rectal therapy to a patient infected with KPC-3 producing *K. pneumoniae*, resulting in complete eradication of the microorganisms without any adverse effects (Corbellino et al. 2020). Another recent study showed that a T₇ like lytic phage of *Podoviridae*, kpsk3, lysed 25 clinical isolates of carbapenem resistant *K. pneumoniae* with a short latent period of 10 min and burst size of 200 PFU/cell (Shi et al. 2020). The phage was able to degrade the exopolysaccharide and did not possess any antibiotic resistant genes or virulent factors in its genome (Shi et al. 2020). Cocktail of phages targeting capsule, membrane proteins and lipopolysaccharide was obtained against KPC producing *K. pneumoniae* strain (Michalik-Provasek et al. 2020). This tri-phage cocktail reduced

the bacterial load by 1–2 log in faecal and caecal samples *in vivo* murine model of GI decolonization (Michalik-Provasek et al. 2020). Lytic bacteriophage belonging to *Myoviridae* from urban sewage, vB_Eco4M-7, was effective against multiple *E. coli* O157 strains, with no toxins and virulence factors (Necel et al. 2020). Another recent study revealed that overexpression of AmpC in *E. coli* promoted its phage lysis, using OmpA (outer membrane protein) as receptor (Wang et al. 2019). Thus increasing ampicillin resistance can enhance its susceptibility to phage, which could pave way for use of antibiotics and phage therapy in synergy. Another study showed that colistin and carbapenem resistant *K. pneumoniae* were highly susceptible to a lytic bacteriophage than its colistin sensitive strain *in vitro*—planktonic, biofilm and *in vivo*—moth larvae and mammalian gut colonization (Hao et al. 2019). The interaction between colistin resistant strain that has less cell surface negative charge and negatively charged phage resulted in low electrostatic repulsion causing the interaction between phage—bacteria and subsequent infection (Hao et al. 2019). Intranasal inoculation of *K. pneumoniae* followed by intraperitoneal administration of phage (SS) within 3 h in a lobar pneumonia mice model, rescued mice from mortality (Chhibber et al. 2008). Cocktail of bacteriophages targeting *E. coli* reduced bacterial load from 2 to 3 log and had less effect on commensal bacteria, relative to ciprofloxacin treatment in an *In vitro* model of small intestine (Cieplak et al. 2018). Another study showed that a phage PDX, belonging to *Myoviridae*, isolated from wastewater in Portland killed diarrhoeagenic enteroaggregative *E. coli* isolates, leaving the human microbiome undisturbed (Cepko et al. 2020). Bacteriophage of *Myoviridae* family, vB_EcoM_JB75 was effective against multidrug resistant *E. coli* isolated from orthopaedic patients (Barros et al. 2019). A T5 bacteriophage, vB_EcoS_AKFV33, was effective against shiga toxigenic *E. coli* with a low multiplicity of infection, it also eradicated the host when mixed with *Salmonella enterica* serovar (Niu et al. 2020). Targeting *E. coli* isolates with dual predators—phage and *Bdellovibrio bacteriovorus* caused a significant reduction relative to individual treatment (Hobley et al. 2020). Phage and *Bdellovibrio bacteriovorus* were able to coexist with each other causing differential pressure on bacterial surface (Hobley et al. 2020).

6. Bacteriotherapy

Bacteriotherapy or faecal microbiota transplantation (FMT) is the introduction of faecal microbiota from a

donor to intestinal tract of a patient to alter the recipient's microbiome composition, thus providing health benefit (Gupta et al. 2016). This therapy has a history from fourth century in treating conditions like diarrhoea to treating pseudomembranous colitis in 1958 and is still continuing (GS B and AJ 1958; Zhang et al. 2012; Manges et al. 2016). A recent study showed that five day treatment with non-absorbable antibiotics (colistin sulphate, neomycin sulphate) followed by FMT from healthy donors decreased ESBL carrying and carbapenem resistant *Enterobacteriaceae* in a randomized clinical trial (Huttner et al. 2019). Another study showed that 20% and 40% of patients ($n = 15$) infected with ESBL producing *Enterobacteriaceae*, tested ESBL negative after first and second faecal transplant, respectively. The recipient's healthy microbiota was restored with a shift towards donor's composition was observed (Singh et al. 2018). An 82-year old female carrying OXA-48 producing *K. pneumoniae* was given a faecal transplant following a bowel lavage (colomycin and gentamicin) (Lagier et al. 2015). Her faeces tested negative for OXA-48 producing *K. pneumoniae* after seven and fourteen days of transplantation, with no adverse effects (Lagier et al. 2015). Another study showed that a 60-year old male with recurrent transplant pyelonephritis caused by ESBL producing *E. coli*, received faeces infusion (Singh et al. 2014). His rectal culture showed ESBL negative after two weeks of transplantation with a twelve-week follow up. He experienced mild diarrhoea and abdominal cramps in first 2 days after infusion, with no further incidences or adverse effects later (Singh et al. 2014). A 14-year old girl was recurrently infected (first infection – 10 months before) with KPC producing *K. pneumoniae* developing osteomyelitis of right femur, and was not cured despite treatment with combination of antibiotics (extended-infusion doripenem, colistin, rifampin, plazomicin) (Freedman and Eppes 2014). She was transplanted with faeces of her younger brother and was advised to receive probiotics for 6 months. Her stool cultures tested negative for KPC and infection did not recur even in 1.5 years after FMT (Freedman and Eppes 2014). In another study, out of eight patients infected with carbapenem resistant *Enterobacteriaceae*, 3 patients were free of CRE colonization after 1 week and 4 were free of CRE three months after FMT, respectively (Dinh et al. 2018). The contribution of human gut phageome in normal intestinal functioning has been emphasized in the recent studies (Townsend et al. 2021). Nevertheless, how different environmental factors and other factors impact the composition of the phageome is still unclear. Antibiotic treatment is likely to cause a decline in gut resident phages, in addition to

the gut bacteria, owing to their non-targeted killing of the bacterial host. Reports also suggest that bacteriotherapy has also been known to restore the phageome and bacteriocins, thus restoring the normal intestinal microenvironment in patients (Zuo et al. 2018).

7. Other mechanisms

Apart from the major alternate therapies discussed earlier, few other approaches have been explored to mitigate these drug resistant pathogens. Most of the approaches involve combinatorial approaches and thus enhancing their efficacies *in toto*. Lavender essential oil (LVO) was found to be synergistic with meropenem and suppressed the growth of carbapenemase producing *K. pneumoniae* (Yang et al. 2020). Mechanistic studies revealed that exposure to LVO caused an increased ROS generation and alteration in cell surface charge and increase in outer membrane permeability thus permeabilizing cell membrane (Yang et al. 2020). Cinnamaldehyde exerted synergistic effect with ciprofloxacin and cefotaxime against ESBL producing quinolone resistant *E. coli* and *K. pneumoniae*, reducing the MIC of the antibiotics by 2–1024 fold (Dhara and Tripathi 2020). The combination altered the expression of genes coding for porins and antibiotic resistance, in addition, it caused damage to bacterial cell structure (Dhara and Tripathi 2020). Linear cationic hexadecapeptide—zp16, exhibited synergy with vancomycin and teicoplanin against carbapenem resistant *K. pneumoniae* and reduced the MIC of vancomycin by >128 fold (Zeng et al. 2020). This remarkable bactericidal activity was due to enhancement of membrane permeability and alteration in membrane potential. Moreover, resistance to the combination was not observed till 20th serial passage (Zeng et al. 2020). Lactoferrin B like peptide (LBLP), derived from centipede *Scolopendrs subpinipes mutilans*, showed antimicrobial activity against pathogenic *E. coli* by causing apoptosis-like death of bacteria due to ROS generation, as evidenced by membrane depolarization, caspase like protein activation, phosphatidylserine exposure and DNA fragmentation (Lee et al. 2020). Capsule depolymerases (Dpo42 and Dpo43) were isolated from IME205 bacteriophage (from sewage sample, active against carbapenem resistant *K. pneumoniae*) (Liu et al. 2020). They effectively made the host bacteria susceptible to serum complement, with no effect on haemolysis of erythrocytes (Liu et al. 2020). Another aspect of targeting pathogenic bacterial strains is by establishing phage antibiotic synergy (PAS) (Comeau et al. 2007). Multiple reports are available that substantiate the enhanced efficiency of PAS against pathogenic

E. coli and *K. pneumoniae* (Tagliaferri et al. 2019). We, in our recent study, have shown that phage KpG, belonging to *Podoviridae*, from the Ganges has shown to exhibit enhanced antibacterial activity causing a 6 log decline in CFU/mL of MDR *K. pneumoniae* in combination with streptomycin *in vitro*, 97% reduction in bio-burden in a zebrafish infection model (Sundaramoorthy et al. 2021). Another recent unpublished study from our group revealed that phage of *Schitoviridae* family from the Ganges river was effective against colistin resistant *E. coli* clinical isolate and had increased activity in combination with sub-MIC concentration of colistin. Sub-lethal doses of ciprofloxacin and ECA2 phage (*Podoviridae*) exhibited synergy against *E. coli*, causing 7.8 log CFU/mL decline in 8 h (Valério et al. 2017). Combination of a β -Lactamase inhibitor (tazobactam) and a membrane permeabilizing antimicrobial peptide (Polymyxin B nonapeptide) potentiated the activity of piperacillin by 128 fold and 64 fold against *E. coli* and *K. pneumoniae*, respectively, *in vitro* (Ferrer-Espada et al. 2020). A dilipid ultrashort tetrabasic peptidomimetic (dUSTBP 8), containing three L-arginine units, potentiated novobiocin and rifampicin against MDR clinical isolates of *Enterobacteriaceae*, owing to its membrane permeabilizing property (Ramirez et al. 2020). A chemical structure designed with a stator and a rotor part acted as a molecular nanomachine, permeabilizing the cell membrane of MDR *K. pneumoniae* upon light activation and allowed enhanced entry of meropenem into the cell, thus increasing intracellular concentration (Galbadage et al. 2019). Colistin was found to exhibit synergy with wide range of antibiotics, especially with protein and RNA synthesis inhibitors against colistin and carbapenem resistant *Enterobacteriaceae* and reduce colonization significantly, exhibiting membrane permeabilizing effect at its sub-MIC (Betts et al. 2014; Brennan-Krohn et al. 2018; Yang et al. 2018). A combination of polymyxin B nonapeptide and vorinostat caused a 5-log decline in CFU/mL of *E. coli* and polymyxin resistant *K. pneumoniae* within 40 min, which was maintained till 24 h, whereas monotherapy of polymyxin B resulted in 2 log decline, which could not prevent the regrowth observed by 24 h (Chen et al. 2020). Another recent study showed that a triple combination (polymyxin B, rifampicin and amikacin) enhanced *in vitro* bacterial killing and caused 1.7 log decline in colony counts in a mouse thigh model infected with polymyxin and carbapenem resistant *K. pneumoniae*, whereas monotherapy with individual antimicrobials was ineffective (Aye et al. 2020). Synergistic combination of fosfomycin-meropenem and fosfomycin-colistin was observed in carbapenem resistant *K. pneumoniae*,

but these combinations failed to maintain significant growth decline in *In vitro* time kill studies (Wang et al. 2018; Bakthavatchalam et al. 2020). Pharmacokinetic MIC modelling revealed that combination of polymyxin B and minocycline can be a potential therapy in treating MDR *K. pneumoniae* (Zhao et al. 2020).

8. Resistance to alternate approaches

Due to evolutionary selection pressures, bacteria are known to develop mechanisms that aid the bacteria to evade killing, especially to antibiotics, within a very short span of time (Baym et al. 2016). There are reports, where bacteria have developed resistance to inhibit the action of adjuvants like enzyme inhibitors, EPIs and in many instances, to bacteriophages. Studies have shown that single substitution in existing β -lactamases can lead to development of novel inhibitor-resistant β -lactamases (Dubois et al. 2004; 2008). Another report showed that the β -lactam inhibitors like clavulanic acid, sulbactam and tazobactam were hydrolysed by KPC-2 β -lactamase (Papp-Wallace et al. 2010). An earlier study showed that the prevalence of resistance to β -lactam inhibitors in *E. coli* isolates from blood samples was due to the production of ESBLs alone and in some cases, it was associated with other mechanisms like porin alteration or hyperexpression of AmpC (Ripoll et al. 2014). A recent study revealed exposure to sub-lethal level of amoxicillin caused modifications in LPS of *E. coli* that in turn led to resistance to phages, which adsorb to receptors on LPS (Zuo et al. 2020). The bacteria were also benefitted with reduction in phage adsorption efficiency by 35% and increased fitness relative to wild type (Zuo et al. 2020). Nevertheless, the reverse strategy, when bacteria develop resistance to phage, it pays high compensatory fitness costs, thus resulting in reduced virulence, decreased incidence of resistance and enhanced antibiotic sensitivity (Burmeister et al. 2020; Hesse et al. 2020). Cocktails of phages targeting different receptors or combination of phages with antibiotics can assure effective killing (Tagliaferri et al. 2019). A plant compound, berberine, has been reported earlier as potent efflux pump inhibitor and is known to sensitize bacteria to aminoglycosides (Morita et al. 2016). It also prevents intestinal adhesion of bacterial pathogens and hence is used as a drug to treat intestinal inflammation. However, lower concentration of berberine (3 & 6 μ g/mL) exerted a hormetic effect by upregulating the expression of KmrA efflux pump in *K. pneumoniae*, and promoted growth in the presence of tetracycline (Li et al. 2021). Reports on resistance to alternate approaches increases the threat of larger bio-burden of drug resistant

pathogenic microorganisms. These studies also underscore the probability of potential risk in using the resistant modulatory compounds as adjuvant therapy. Thus, detailed and thorough studies on all aspects are warranted before these compounds are taken up for translational purposes.

9. Conclusion

Evolution of resistance to antibiotics by bacteria, especially, *Enterobacteriaceae*, is of greatest concern, as they are one of the predominant causes of life threatening infections. Among them, *E. coli* and *K. pneumoniae* are the most dreadful owing to their fast emerging antibiotic resistance, even to the last resort antibiotics like colistin. Development of new antibiotics may be effective, but resistance to it can arise (evolve) within a short time, moreover the antibiotic development process costs more time and requires exorbitant monetary investment. Hence, non-conventional approaches that target the resistance mechanisms of bacteria like inhibition of efflux pumps, biofilm inhibition, curing of plasmids to prevent and control spread of AMR genes and inhibiting essential enzymes for bacterial survival can be efficient and wise mode to tackle the MDR strains. In addition, bacteriophages targeting pathogenic strains can also help widely in mitigating the resistant infections. These can be given as adjuvant therapy along with existing antibiotics, which would provide a dual attack against the pathogens. In retrospective, analysis of long-term effects on usage of the adjuvants has to be carefully studied, to avoid adverse reactions. Bacterial resistance to the alternate therapies might evolve, which can be prevented by using a combination therapy. Despite plenty of research publications available to combat MDR pathogens, most of the discussed approaches have been shown to be feasible in laboratory conditions and not at clinic (bedside). Realizing the urgent need to deal with the pathogen from critical priority list, the effective combinations should be taken to clinical trials and has to be made available for medical emergencies, without any further delay, thus saving multiple lives from the life threatening infections.

Disclosure statement

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